PYTHON FUNDAMENTALS

ASSIGNMENT 01

Beginner Practice
Input · Types · Operators · Formulas

YOUR PRACTICE ROADMAP

01 · Input & Output Read user input and display clear results.	**02 · Data Types** Convert between strings, integers and floats.
03 · Operators Use arithmetic and understand precedence.	**04 · Formula-Based Problems** Apply formulas to everyday calculations.

How to use this assignment
Solve each problem yourself first. Use meaningful variable names, add short comments where helpful, and test your program with more than one input. Do not worry about writing the shortest code—the goal is to understand each concept.

PART A · ORIGINAL ASSIGNMENT

QUESTION 01 · Personal Greeting

Write a program that asks the user for their name and age, then prints a sentence like:

```
Hello Shradha, you are 21 years old!
```

Practice focus: input(), variables, f-strings

QUESTION 02 · Basic Calculator

Take two numbers as input and print their sum, difference, product, and quotient.

Practice focus: numeric conversion, arithmetic operators

Hint: Convert the inputs to a numeric type before performing calculations. Consider what happens if the second number is zero.

QUESTION 03 · Average of Three Values

Ask the user to enter two integers and one float. Convert all three values to floats and print their average.

Practice focus: float(), arithmetic, division

QUESTION 04 · Type Conversion

The user enters a string containing a number (for example, `"45"`). Convert it to an integer, a float, and a string again. Print each value and its type.

Practice focus: int(), float(), str(), type()

QUESTION 05 · Operator Precedence

Evaluate and print the result of this expression:

```
x = 10 + 3 * 2 ** 2
```

Then explain why Python produces that result.

Practice focus: operator precedence: exponentiation, multiplication, addition

PART A · ORIGINAL ASSIGNMENT (CONTINUED)

QUESTION 06 · Swap Two Values

Ask the user for two numbers and swap their values. Print the values before and after swapping.

Practice focus: variables, assignment

QUESTION 07 · Celsius to Fahrenheit

Ask for a temperature in Celsius as text input, convert it to a float, and print the temperature in Fahrenheit.
Formula: Fahrenheit = (Celsius × 9/5) + 32

Practice focus: input conversion, arithmetic

QUESTION 08 · Circle Area

Ask for the radius of a circle and calculate its area using $\pi = 3.14$.
Formula: Area = 3.14 × radius²

Practice focus: float(), exponentiation, constants

QUESTION 09 · Simple Interest

Ask for principal (P), rate (R), and time (T). Convert all inputs to floats and calculate simple interest.
Formula: SI = (P × R × T) / 100

Practice focus: multiple inputs, numeric conversion

QUESTION 10 · Integer and Fractional Parts

Take a decimal number such as 45.78 and display its integer part (45) and fractional part (0.78). Decide how your solution should handle negative values.

Practice focus: int(), subtraction, floating-point behaviour

Hint: For this beginner exercise, you may start with positive values and then test a negative number.

PART B · EXTRA PRACTICE

These additional questions extend the same fundamentals. Try them after completing the original ten.

QUESTION 11 · Minutes to Hours and Minutes

Ask the user for a total number of minutes. Display the equivalent number of whole hours and remaining minutes.

Practice focus: integer division (//), remainder (%)

QUESTION 12 · Shopping Bill

Ask for the price of one item and the quantity purchased. Calculate the subtotal, 5% tax, and final bill.

Practice focus: float(), multiplication, percentages

QUESTION 13 · Last Digit

Ask the user for a positive integer and print its last digit without converting it to a string.

Practice focus: remainder operator (%)

QUESTION 14 · Rectangle Calculator

Ask for a rectangle's length and width. Print its area and perimeter.

Practice focus: Area = length × width; Perimeter = 2 × (length + width)

QUESTION 15 · Average Speed

Ask for a distance in kilometres and a travel time in hours. Calculate average speed in km/h. Handle a time of zero sensibly.

Practice focus: division, input validation

QUESTION 16 · Seconds Converter

Ask for a total number of seconds and display hours, minutes, and remaining seconds.

Practice focus: // and %

QUESTION 17 · Two-Digit Number

Ask for a positive two-digit integer. Print its tens digit and ones digit using arithmetic only.

Practice focus: integer division (//), remainder (%)

QUESTION 18 · Percentage Calculator

Ask for marks obtained and total marks. Calculate and print the percentage. Avoid division by zero.

Practice focus: division, multiplication, validation

QUESTION 19 · Swap Without a Third Variable

Ask for two numeric values and swap them without creating a third variable. Explain how your approach works.

Practice focus: assignment, arithmetic or Python tuple unpacking

QUESTION 20 · Expression Challenge

Ask the user to enter values for a and b, then calculate and print $(a + b)^2$ and $a^2 + b^2$ separately. Explain why these expressions are generally different.

Practice focus: operator precedence, exponentiation, algebra

SUBMISSION & SELF-CHECK

Before committing to GitHub

- ■ Each question is solved in its own clearly named .py file, or in one notebook with numbered sections.
- ■ Variable names are readable and code is consistently formatted.
- ■ Programs are tested with normal and edge-case inputs (for example, zero where division is involved).
- ■ No solutions are copied without understanding them.
- ■ Add a short README explaining the topics practised and how to run the solutions.

Learning note: This worksheet contains problem statements, not solutions. Attempt the problems independently, then review and improve your code.